

Sharp Matrix-Scaling Stability over Surreal and Surcomplex Fields

Spanning-tree gaps, infinitesimal normalization,
and exact multiscale propagation

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Abstract

We study diagonal matrix scaling in set-sized Hahn fields $F = \mathbb{R}((t^\Gamma))$ and $K = \mathbb{C}((t^\Gamma))$, with no rank or divisibility assumption on the ordered value group for the local theory. A positive matrix may have entries at arbitrarily different infinitesimal scales. Nevertheless, normalization to its fixed row and column sums admits a unique support-controlled surcomplex analytic branch on the entire relative infinitesimal polydisc.

The main quantitative invariant is the *spanning-tree deletion gap*. For an edge e of the bipartite support graph, let τ be the minimum sum of entry valuations over spanning trees and let τ_e^- be the same minimum over trees avoiding e . Then $\kappa_e = \tau_e^- - \tau$ is the uniform gain in relative valuation at entry e , and is optimal when $\Gamma \neq \{0\}$: an input change of valuation at least $\delta > 0$ produces an output change of valuation at least $\delta + \kappa_e$. The assertion is a nonlinear two-point estimate, not merely an infinitesimal derivative formula. Every change in just entry e attains equality at that entry; a bridge is fixed exactly. All positive-degree Taylor coefficients satisfy the same entrywise valuation bounds.

A tree interpolation identity and a secant reweighting of the exponential prove the sharp estimate. A separate finite-generator support argument licenses evaluation at arbitrary rank, where integrality of formal coefficients alone is insufficient. For a tridiagonal bistochastic chain we obtain an exact nonlinear propagation law: valuation gains add along consecutive links. We also give a real-linear-constraint extension, an unramified local descent statement, explicit examples, and exact finite verification programs.

Status and priority. The main theorems are proved in this manuscript. Their originality is a *candidate* conclusion of a targeted, non-exhaustive comparison with published work and the repository at commit 048b72c. No named long-standing open problem is claimed solved, and no refereeing or complete machine verification is claimed. Known scaling, projection, and spanning-tree results are explicitly credited. This is an AI-assisted research draft. Its 16 standard statements are inventoried in the collection's ledger, but none has an Implementation mapping; no Lean proof coverage is claimed here. It is built from a single manuscript; [section 1.7](#) records how that manuscript was placed in the collection and what was changed.

Contents

1	The problem and the contribution	1
1.1	Why this is more than scalar extension	1
1.2	An overview of the exact result	1
1.3	What is proposed as new, and what is not	2
1.4	What this report does not claim	2
1.5	Relation to other reports of the collection	4
1.6	Words and symbols used differently elsewhere in the collection	5
1.7	Provenance of this report	6
2	Literature and repository comparison	7
2.1	The closest published precedents	7
2.2	Comparison with the specified repository	8
3	Hahn conventions and the summability issue	9
3.1	Fields, valuation, and positive leading coefficients	9
3.2	Strong summability	9
3.3	Restricted exponential and logarithm	11
4	The tree interpolation identity	11
4.1	Incidence, cuts, and weighted cycles	11
4.2	The row formula and the diagonal coefficient	12
5	Normalization on the whole infinitesimal polydisc	13
5.1	The nonlinear equations	13
5.2	Gauge factors, retraction, and local coordinates	14
6	Sharp nonlinear valuation gains	15
6.1	Secant reweighting	15
6.2	The optimal two-point bound	15
6.3	All Taylor orders and a remainder certificate	17
7	Computing the gain without summing trees	18
8	Exact nonlinear propagation through a chain	19
8.1	The matrix and the response	19
8.2	Cycle coordinates and the secant matrix	19
8.3	A rank-two example	21
9	Global positive matrix scaling and its relation to the branch	21
9.1	Classical existence and real-closed transfer	21
9.2	An elementary multiplicative comparison	22
10	Extension to real linear constraints	23

11 Surreal specialization and preservation of the support group	24
11.1 No ramification in the relative infinitesimal problem	24
11.2 Transport to surreal and surcomplex numbers	24
12 Examples, boundaries, and failure modes	25
12.1 A closed-form square example	25
12.2 Positivity cannot simply be removed	26
12.3 Fixed margins are essential	26
12.4 No topological convergence claim is hidden in the proof	27
13 Verification, implementation, and remaining questions	27
13.1 What the supplied program checks	27
13.2 A reproducible symbolic workflow	28
13.3 Questions not settled here	28
14 Conclusion	28
A Proof dependency map	29
B Build and artifact record	29

1 The problem and the contribution

1.1 Why this is more than scalar extension

Let $P = (p_{ij})$ be a finite matrix with positive surreal entries on a fixed support. Change its nonzero entries by relative infinitesimals, and then multiply rows and columns by positive factors so that the original margins are restored. How accurately are the individual entries of the normalized matrix determined?

A direct inverse-Jacobian bound can suggest a loss of precision when some entries are extremely small. A different phenomenon occurs for fixed margins. The normalized entries never lose relative valuation, and certain entries gain a precisely computable additional amount. The answer depends on which edges are indispensable to minimum-valuation spanning trees, rather than on a smallest eigenvalue used as a scalar condition number.

This is a genuinely multiscale question. For example, in $\Gamma = \mathbb{Z}^2$ with lexicographic order, put $\alpha = (0, 1)$ and $\beta = (1, 0)$. Then $n\alpha < \beta$ for every ordinary positive integer n . An argument requiring $n\alpha$ to become arbitrarily large cannot justify a Hahn expansion at t^α . The proofs below retain the entire ordered group and use strong summability, not an implicit replacement by a real-valued norm.

The word *surcomplex* means the complexification $\mathbf{No}[i]$. For rigorous constructions we first work in the set-sized field $K = \mathbb{C}((t^\Gamma))$. The final passage to surreal and surcomplex numbers uses the normal-form embedding and introduces no class-sized sums; see [section 11](#). The local existence theorem is actually slightly stronger than a real-positive statement: each base weight may be surcomplex, provided its leading coefficient is positive real.

1.2 An overview of the exact result

For a connected bipartite graph $\mathcal{G} = (R \sqcup C, E)$ and edge weights p_e , write $c_e = v(p_e)$. Let $\mathcal{T}$ be its finite set of spanning trees. Define

$$\tau = \min_{T \in \mathcal{T}} \sum_{a \in T} c_a, \quad \tau_e^- = \min_{\substack{T \in \mathcal{T} \\ e \notin T}} \sum_{a \in T} c_a, \quad \kappa_e = \tau_e^- - \tau. \quad (1.1)$$

An empty minimum is $+\infty$. Thus a bridge has $\kappa_e = +\infty$. For $f \in \mathbf{m}_K^E$ let the perturbed kernel be $A_e(f) = p_e \exp(f_e)$, using only the infinitesimal exponential. Let $q(f)$ denote its normalization to the margins of p on the relative infinitesimal branch. The central estimate is

$$v\left(\frac{q_e(f)}{q_e(g)} - 1\right) \geq \kappa_e + \min_{a \in E} v(f_a - g_a), \quad f, g \in \mathbf{m}_K^E. \quad (1.2)$$

When $f - g = se_e$, where $0 \neq s \in \mathbf{m}_K$, equality holds in (1.2) for a nonbridge e . If $\Gamma \neq \{0\}$, such an s exists (take $s = t^\gamma$ with $\gamma > 0$), so no larger uniform gain can replace κ_e . When $\Gamma = \{0\}$, the infinitesimal domain is the singleton $\{0\}$; the coefficientwise sharpness below still holds, but evaluated optimality has no witness. The equality for nonzero changes holds even when minimum trees are tied and when the perturbations have nonreal coefficients.

An equivalent combinatorial formula needs no tree enumeration. If e joins u to v , then

$$\kappa_e = \max \left\{ 0, \min_{\pi: u \rightsquigarrow v \text{ in } \mathcal{G} \setminus e} \max_{a \in \pi} c_a - c_e \right\}. \quad (1.3)$$

The value is $+\infty$ if no alternative path exists. The minimax path formula is familiar minimum-spanning-tree sensitivity. Its role as the *optimal nonlinear normalization gain*, including all Taylor orders, is the contribution investigated here.

1.3 What is proposed as new, and what is not

Result	Status in this manuscript
Scaling existence over real closed fields	Classical real scaling plus first-order transfer; included as background.
Weighted constraint projection and tree formulas	Known linear-algebraic and transfer-current ingredients; proved directly for transparency.
Whole-infinitesimal-polydisc normalization	Candidate original arbitrary-rank, positive-leading surcomplex theorem, with an explicit strong-summability proof.
Sharp tree-deletion gain	Candidate original nonlinear two-point law, coefficient bounds, and exact one-coordinate sharpness.
Propagation through a chain	Candidate original exact nonlinear valuation formulas at all links and diagonal entries.
Real-linear-constraint extension	A proved extension replacing trees by bases; no new matroid combinatorics is claimed.

The first derivative alone is *not* a novelty claim. Nor is the mere existence of generalized series for a one-parameter scaling problem. Both have substantial precedents, described next.

1.4 What this report does not claim

Each limitation below is also stated where it applies. They are collected here so that none of them is lost when the results are quoted elsewhere.

- (N1) *Priority and status.* The whole-polydisc normalization, the sharp tree- and basis-gap laws and the chain propagation formulas are *candidate* original results after a targeted, non-exhaustive comparison with published work and with the repository at its pin (section 2). Publication priority, absence from every repository file and independent expert validation are not certified. The repository audit was not a line-by-line review of every article or Lean declaration, and its indexed searches were unusable as negative evidence. No named long-standing open problem is claimed solved. The report is an AI-assisted draft; it has not been refereed; none of its results has a Lean Implementation mapping in the ledger, and section 13 does not assert that the Lean modules it outlines exist.
- (N2) *Credited, not claimed.* Existence, uniqueness and ordinary real continuity of prescribed-margin scaling (Idel’s review and classical scaling theory); generalized-series expansions of a one-parameter scaling problem, so that no first use of Hahn-type series in matrix scaling is claimed (Sharify–Gaubert–Grigori); the first-order implicit differentiation formula (Eisenberger et al.); the determinant formula, its tree interpretation and transfer-current representations (Burton–Pemantle); the minimax path formula (1.3) and the minimum-spanning-tree exchange arguments of section 7; Hahn and surreal normal-form background (Gonshor, Berarducci–Mantova); Higman’s word theorem; real-closed transfer (Tarski). The real-linear-constraint extension (theorem 10.1) claims no new matroid combinatorics.
- (N3) *No convergence assertion.* Neither convergence of the alternating Sinkhorn iteration in an arbitrary-rank valuation topology, nor its iteration complexity, nor convergence of set-indexed sequences in the full surreal fine topology is claimed. Strong sums are

formed coefficientwise; strong summability alone does not imply convergence of partial sums (section 12). The remainder certificate (corollary 6.5) does not promise that a finite Taylor degree reaches every requested precision at higher rank.

- (N4) *What “analytic” means.* The normalization branch is given by a strongly evaluated formal series on the relative infinitesimal polydisc. No classical holomorphic manifold structure in the full surreal fine topology is asserted. No exponential or logarithm at an infinite surcomplex argument, no global surcomplex exponential and no surreal derivation is used; Taylor coefficients are formal derivatives with respect to the edge parameters.
- (N5) *The hypotheses matter.* Positive leading coefficients of the base weights are sufficient, not claimed to be necessary; without them integrality and uniqueness can fail (examples 12.1 and 12.2). The field K is not ordered, and “positive-leading” refers only to the leading coefficient. The margins are fixed; if they move, even by relative infinitesimals, the bridge rigidity fails (section 12). The support is fixed.
- (N6) *What “sharp” means.* For $\Gamma \neq \{0\}$, κ_e is the largest gain valid uniformly over infinitesimal inputs. Formal coefficient sharpness also holds for the trivial group; equality is not asserted for every input direction (theorem 6.2 and the remark after it). Optimality of the exponent N in proposition 9.2 is not claimed.
- (N7) *Which costs.* The tree costs of the sharp theorem are valuations of the normalized base matrix. No equally sharp formula from the valuations of an unscaled kernel is claimed (corollary 9.3).
- (N8) *Divisibility and real closedness.* The local theory assumes no divisibility of Γ . The global positive interpretation (proposition 9.1 and corollary 9.3 and the global paragraph of section 11) uses real closedness, reached through the divisible hull and first-order transfer; the real optimization proof is not repeated inside a non-Archimedean field, and no non-Archimedean compactness or global logarithm is used. The descent statement corollary 11.1 is local: global scaling over a nondivisible group can require ramification (11.1).
- (N9) *Real constant constraints.* Theorem 10.1 needs a real constant constraint matrix. The same basis-cost formula is not asserted for a Hahn-valued constraint matrix.
- (N10) *Surreal transport.* The surreal and surcomplex statements are obtained by set-sized support-group localization. No class-sized sum, net or compactness argument is used, and no proof of a local result uses spherical completeness or Banach contraction.
- (N11) *Computation.* The workflow of section 13 is an exact-operation specification, not a universal computable representation of surreal numbers: arbitrary Hahn coefficients need not come with an effective leading exponent, equality test or support oracle. The amplitude C_e needs all minimizing trees when there are ties. The finite checks of section 13 do not prove strong summability, arbitrary-rank existence or surreal transport, and a successful compilation is not a proof check.
- (N12) *Open questions.* The three questions of section 13 (a sharp two-edge propagation formula on a general support graph, of which the chain theorem solves one family; a classification of the surcomplex base weights with a unique integral whole-polydisc branch; and the convergence and complexity of alternating normalization over higher-rank valued fields, together with a global surcomplex scaling theorem for arbitrary complex kernels) are not settled.

The delivered files `SOURCE_AUDIT.md` and `PROOF_AUDIT.md` state the audit boundaries in more detail.

1.5 Relation to other reports of the collection

The proofs in this report are given directly; the neighbouring reports below supply comparisons. These were added during editorial assembly at `ed88b8f` and checked again against the source labels at `034ab96` during the main-text review. Historical search claims remain tied to their snapshots in [section 2](#).

Markov generators at every scale. The Markov report [\[10\]](#), held in `docs/surreal/markov-generators-at-every-scale/`, and this one share one mechanism: a positive combinatorial expansion whose valuation is a minimum over trees or forests and whose leading coefficient is a positive sum over the minimizers. There, Proposition 3.1 (`markov:prop:forest`, the directed matrix-forest theorem) writes $\det(sI + L)$ and $\text{adj}(sI + L)$ as sums of positive in-forest weights, and equation (3.10) there (`markov:eq:profile-properties`) reads off the minimal forest weight and its amplitude. Its Theorem 4.1 (`markov:thm:leading`) gives the valuation of a resolvent entry as a minimum over forests with a prescribed root, minus the unrestricted minimum. Here [theorem 4.1](#) writes $\det L_p$ as the positive tree sum $Z(p)$, [\(4.6\)](#) writes a diagonal projection entry as $Z_e^-(p)/Z(p)$, and the gap $\kappa_e = \tau_e^- - \tau$ is again a restricted minimum (trees avoiding e) minus the unrestricted one, with the quotient of amplitudes C_e in [\(6.3\)](#). As polynomial identities in the weights, symmetric rates $q_{uv} = q_{vu} = p_e$ on the support graph make the in-forests of that report with $N - 1$ edges and a fixed root exactly the spanning trees oriented towards that root, so its top forest coefficient σ_{N-1} equals $N Z(p)$. This report does not use that identity.

The differences are as follows. (i) *Objects*. There, the row Laplacian of a directed graph, in general neither symmetric nor normal, and the normalized resolvent, a rational function of the rates and of a scale parameter, with forests of every size. Here, the symmetric reduced Laplacian $BW_p B^\top$ of a bipartite support graph, only spanning trees, no scale parameter, and a *nonlinear* fixed-margin normalization obtained from a support-controlled power series ([theorem 5.1](#)). (ii) *Stability*. Theorem 8.1 there (`markov:thm:stability`) turns relative rate errors of valuation at least δ into relative resolvent-entry errors of valuation at least δ , uniformly in the resolvent parameter, and Example 8.3 there shows that no gain is possible in general. [Theorem 6.2](#) turns relative kernel changes of valuation at least δ into normalized-entry changes of valuation at least $\delta + \kappa_e$, with equality for one-coordinate changes; the gain comes from the fixed margins and fails when the margins move ([section 12](#)). The two theorems concern different maps, and neither implies the other. Both need a fixed support: a new edge is not a relative perturbation (Remark 8.4 there). (iii) *Hypotheses*. That report assumes a divisible value group throughout and real positive rates. The local theory here assumes no divisibility and allows surcomplex positive-leading base weights with complex perturbations. (iv) The crossover hierarchy, effective generators, rank-one realization and eigenvalue amplitudes of that report have no counterpart here. The whole-polydisc branch, the deletion gap and its bottleneck formula, and the chain propagation law of this report have no counterpart there.

Surcomplex spectral theory. The spectral report `docs/surcomplex/spectral-theory/` [\[11\]](#) shares three ingredients with this one.

- (a) *Positive Cauchy–Binet sums*. Its Theorem 8.1 (`thm:gram`, a label without report prefix) writes the coefficients of $\det(I + sA^*A)$ as sums of squared moduli of minors, so that no leading cancellation occurs. [Theorem 4.1](#) expands $\det(BW_p B^\top)$ by Cauchy–Binet as

$\sum_T (\det B_T)^2 w_T(p)$, and [theorem 10.1](#) has the same form with real factors $(\det B_T)^2 > 0$. When the weights are real and positive and have square roots in the field, $BW_p B^\top = A^* A$ with $A = W_p^{1/2} B^\top$, and $\det L_p$ is the top Gram coefficient of that theorem. The proof here needs neither square roots, which a nondivisible value group need not supply, nor real weights: it applies Cauchy–Binet to the product $B \cdot W_p \cdot B^\top$ with positive-leading surcomplex weights. The interpolation identity (4.3) for the projection itself has no counterpart there.

- (b) *The support engine.* [Lemma 3.3](#) and [Lemma 9.4](#) there (`spec:lem:neumann`) state the same positive-support word finiteness, both from Higman’s theorem. [Lemma 3.4](#) with no generators ($R_0 = \mathbb{C}$) gives the strong-evaluation part of [Lemma 9.5](#) there (`spec:lem:evaluate`); [lemma 3.4](#) allows coefficients in a finitely generated subalgebra of $\mathcal{O}$, and [example 3.5](#) shows why arbitrary coefficients in $\mathcal{O}$ are not allowed. [Example 9.2](#) there (`spec:ex:nonconvergence`) and the last subsection of [section 12](#) give the same failure of partial-sum convergence, with the elements $(0, 1)$ and $(1, 0)$ of $\mathbb{Z}_{\text{lex}}^2$. Strong summability alone does not imply such convergence. Each report keeps its own proofs.
- (c) *No new exponents.* Its [Theorem 9.18](#) (`spec:thm:svd`) keeps singular value decompositions in the subgroup generated by the input supports, as [corollary 11.1](#) keeps the local normalization in $\mathbb{C}((t^\Lambda))$. Its [Section 10](#) computes when positive matrix roots need new exponents. The global ramification example (11.1) here is computed directly and is not derived from that section.

What is not shared: that report is about eigenvalues, singular values and Hermitian or normal matrices, and its perturbation results (for instance [Theorem 5.2](#) there, `thm:weyl`) bound changes of eigenvalues and singular values. No eigenvalue or singular value appears in this report. The gain κ_e is a combinatorial invariant of the support graph and the entry valuations, not a smallest singular value used as a condition number ([section 1](#)), and singular-value stability supplies neither the whole-polydisc support argument nor the tree-deletion sharpness ([section 2](#)).

Prony reconstruction. The nonuniform certificate of the Prony reconstruction report ([Theorem 8.1](#) there, `prony:thm:graph`) is also a minimum-cost path computation in an ordered group. Its graph is a directed graph of moment interactions. Its sufficient reconstruction certificate requires positive cycle weights and the strict-ball inequalities $\rho_i - E_i > \delta_i$ for every i ; its path minima then bound node errors. Those letters are local to that report. It involves no spanning trees and no normalization, and its moment precisions κ_k are unrelated to the gaps κ_e .

1.6 Words and symbols used differently elsewhere in the collection

Several words and letters of this report mean other things in other reports. Each has one meaning here.

- *Scaling* and *normalization* mean diagonal row and column scaling of a matrix to prescribed margins, and its local branch $\mathcal{N}_p, q_p$. They are not the monomial rescalings or polynomial normalizations of other reports. *Scale*, as in the title, means a valuation.
- *Positive*, for a matrix, means entries positive on the support in the ordered field F ; *positive-leading* ([definition 3.1](#)) is the only positivity notion in K , which is not ordered. Neither means positive definite, the sense of the spectral report.
- *Gap* means the spanning-tree deletion gap κ_e . It is not a spectral gap and not a surreal

gap (a Dedekind section of **No**) in the sense of the genetic-gap report. The letter κ also denotes moment precisions in the Prony report and the valuation gain $v(q)$ of an inverse branch in the expanding-polynomial-dynamics report; those are unrelated.

- *Tree* means an undirected spanning tree of the support graph. It is not an in-forest of the Markov report, nor a cluster, separation or splitting tree of the polynomial, Prony or spectral reports.
- *Branch* means the unique relative-infinitesimal solution $\mathcal{N}_p$ on $\mathfrak{m}^E$, and *analytic* means given there by a strongly evaluated formal series with coefficients in R_p . These are not the germ-analytic or coherent-section classes of the analysis report, and the branch is not a branch selector in the computational sense of `docs/NOTATION.md`.
- *Gauge factors* are the diagonal row and column factors of [corollary 5.2](#); neither the noise gauge of the Prony report nor a differential gauge equivalence.
- *Bistochastic* means that all row and column sums are 1.
- p denotes base weights and $q_p(f)$ normalized weights; P and Q are the corresponding matrices. In the Markov report q_e are rates and P_i idempotents. $L_p = BW_p B^\top$ is a symmetric reduced weighted Laplacian, not that report's row Laplacian. $H_p = I - \Pi_p$ is a projection, not a Hermitian matrix H as in the spectral report, and Π_p is unrelated to the exponent matrix Π of the theta-series material in the Hahn–Tate report. s is a perturbation scalar, not a resolvent parameter.
- Local letters are reused with local meanings, each defined where it is used: δ (the least valuation of an input change in [sections 1, 6 and 8](#), the group element $(0, 1)$ in [example 3.5](#) and [section 12](#), and a positive parameter in the examples of [section 12](#)); C (the column set, the Cramer matrices C_k , the amplitude C_e , the cycle basis of [section 8](#)); R (the row set, $R(z) = \exp z - 1 - z$, the rings R_0 and R_p); D (the diagonal factors D_r, D_e , the continuants $D_{a,b}$, a truncation degree). The formal series of [lemma 3.4](#) and [example 3.5](#) is written $\Phi(X)$, so that F always denotes the real Hahn field (F_0 in [proposition 9.1](#) is an arbitrary real closed field).
- The valuation is $v(x) = \min \text{supp } x$, and the surreal embedding is $t^\gamma \mapsto \omega^{-\gamma}$ ([11.2](#)), as elsewhere in the collection. *Surcomplex* means **No**[i].

1.7 Provenance of this report

This report is built from one manuscript, *Sharp Matrix-Scaling Stability over Surreal and Surcomplex Fields*, dated 22 September 2026. It arrived as the archive `surreal_matrix_scaling.zip`, one of nine archives committed on arrival at `330aa79`. The archive held the LaTeX source `surreal_matrix_scaling.tex`, its PDF, a README, the audit files `SOURCE_AUDIT.md` and `PROOF_AUDIT.md`, the program `code/verify.py` with its recorded output and console record, a build record with its console log, a requirements file and a build script. It was placed as raw source at commit `30dfb4f`; editorial assembly at `ed88b8f` added the report PDF and the material described below. Its own repository comparison is pinned to `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0` ([section 2](#)).

There is no second source, so there was no merge and no choice between sources. Nothing was selected out of a larger body of work. Every theorem, proof, example and disclaimer of the manuscript is kept. The report is a member of the surreal family because its positive matrices, the global scaling theory and the tree costs are real Hahn objects; the surcomplex branch of [theorem 5.1](#) extends that real theory.

Raw placement and subsequent editorial assembly made these changes.

- (a) On raw placement, the source was renamed `article.tex`. The two audit files, the program and everything in `data/` are kept byte-identical. The build script was moved to `code/` and the requirements file to `data/`. The delivered PDF and README are not shipped.
- (b) All 88 labels of the manuscript already carried the prefix `scale:` and are unchanged. The labels added during assembly carry the same prefix.
- (c) Sections 1.4 to 1.7 were added during assembly, and so were remarks 3.6 and 4.2, which point to other reports. They were placed so that every theorem, equation and section number of the manuscript is unchanged. The title-page status paragraph now states the draft status.
- (d) Repository statements that had gone stale were corrected, with the pin kept as provenance (the paragraph “Since the pin” in section 2, and appendix B, whose build instructions now describe this directory). Section 13 records a rerun of the checks during assembly.
- (e) The formal series written $F(X)$ in lemma 3.4 and example 3.5 was renamed $\Phi(X)$, since F is the real Hahn field.
- (f) All numbered statements share one counter. In the delivered build, every cross-reference to a proposition, lemma, corollary, definition, example or remark printed the word “theorem”. Six lines in the preamble now make each reference print the right name; no number changed.

These assembly changes left the mathematics unchanged. The subsequent main-text review of Sections 1–14 expanded proofs and corrected the scope and comparisons: it distinguished evaluated sharpness from formal coefficient sharpness at the trivial group, specified strict feasibility on the whole support, corrected two published theorem citations, and separated strong summability from partial-sum convergence. The delivered audit, code and data remain unchanged. The review is mathematical reading, not Lean coverage or a literature-priority certificate; the collection’s `docs/REVIEW.md` records its scope.

2 Literature and repository comparison

2.1 The closest published precedents

The prescribed-margin matrix-scaling problem and its support conditions are classical. Idel’s review [6] is a useful entry point; its Theorem 3.1 gives a general scaling criterion, and Theorem 4.5 (p. 26 of the cited v1) gives continuity on the fixed prescribed-margin pattern class. The present article does not reclaim existence, uniqueness, or ordinary real continuity.

Sharify, Gaubert and Grigori [7], Theorem 2.4 (p. 6 of v2), prove that for a nonnegative square matrix A with total support, the bistochastic scaling of its Hadamard deformation $A^{(p)}$ has absolutely convergent generalized Dirichlet expansions in $t = \exp(-p)$. Their Theorem 2.3 gives convergence to the unique maximum-entropy point of the optimal-assignment face. This is an important prior connection between matrix scaling, generalized series, and tropical methods. Our question is different: we fix an arbitrary-rank Hahn matrix, allow arbitrary strongly defined relative infinitesimal changes, and identify the exact edgewise nonlinear valuation gain. We make no claim to the first use of Hahn-type series in matrix scaling.

Eisenberger, Toker, Leal-Taixé, Bernard and Cremers [3] derive an implicit differentiation framework for Sinkhorn normalization. The inverse weighted constraint matrix appearing there is the same linear-algebraic object that underlies our projection. Their work also analyzes

approximate inputs. Our additional conclusions concern exact nonlinear changes in an ordered value group, integrality of all Taylor coefficients, and support-controlled evaluation beyond rank one.

Spanning-tree representations of electrical projections and transfer-current matrices belong to a well-developed theory; see Burton and Pemantle [2]. We use an elementary Cauchy–Binet and Cramer’s-rule proof of the particular interpolation identity needed here. The determinant formula, its tree interpretation, and ordinary minimum-spanning-tree exchange arguments are not proposed as new.

The Hahn-field and surreal background is standard; the normal-form conventions can be compared with Gonshor [4] and the preliminaries of Berarducci and Mantova [1]. The support lemma used below is proved from the standard word-ordering theorem of Higman [5]. None of these foundational ingredients is claimed as a contribution of this article.

2.2 Comparison with the specified repository

The audit is pinned to the repository commit

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0.

The complete research catalogue `docs/README.md` at that pin lists 36 reports. The nearest subjects are the Markov-generator report and the spectral-theory report [9, 10, 11]. The former studies normalized resolvents, positive forest formulas, scale hierarchies, and relative stability of resolvent entries. The latter develops finite spectral theory, valuation scales, positive Cauchy–Binet identities, and exact field-of-definition questions.

Our shared mechanism with those reports is the absence of leading cancellation in positive finite sums. Our object is different: a nonlinear fixed-margin diagonal normalization, its surcomplex local branch, and its optimal entrywise gains. A stability theorem for $s(sI + L)^{-1}$ is not the normalization theorem proved below. Likewise, singular-value stability does not provide the whole-polydisc support argument or the tree-deletion sharpness statement.

The audit read the catalogue, the Markov report’s detailed guide, and the first 180 lines of the spectral report’s guide. The directory `docs/new` at the pin contains only its README, not unexamined ZIP archives. These observations support choosing the present topic; they are not a line-by-line audit of every article or Lean declaration. In particular, indexed connector searches were unusable as negative evidence: searches for `Sinkhorn` and `spanning` returned no results, but so did the positive-control query `MarkovResolvent`, a file name present in the repository tree.

The accompanying `SOURCE_AUDIT.md` records the scope and search phrases. Searches that returned irrelevant results were not treated as evidence of absence. The appropriate priority claim is therefore: *the main formulations were not found in the sources actually compared, and are submitted as candidate original results*. There is no assertion that every publication or every repository line has been excluded as a possible precedent.

Since the pin. The manuscript was placed as raw source at commit `30dfb4f`. The statements above describe the pin and are kept as provenance. This paragraph was added during editorial assembly at `ed88b8f`; its catalogue and search observations describe the raw-placement snapshot. The catalogue `docs/README.md` did not change between the pin and raw placement: it listed 36 reports, excluding this report and the three others placed with it. The directory `docs/new` held only its README at the pin, as stated above. Commit `330aa79` then added nine delivered archives, this manuscript’s among them; commit `30dfb4f` unpacked, placed and removed all nine, so that snapshot again held only the README in this directory. The

positive-control file `Surreal/Algebra/MarkovResolvent.lean` was still present. It proves finite resolvent and projection-flag algebra for the Markov report and contains no spanning-tree, forest-sum or Cauchy–Binet statement. At 30dfb4f, a full-text search of every LaTeX source in the collection finds the words `Sinkhorn`, `bistochastic`, `Kirchhoff` and `transfer current` only in this report. The phrase `spanning tree` occurs elsewhere only once, in the proof of the Markov report’s matrix-forest identity [10], where it means a directed spanning tree rooted at a vertex. The phrase `matrix scaling` occurs elsewhere only once, in the outline of remaining Lean development of the Prony reconstruction report (`docs/surcomplex/prony-reconstruction-at-surreal-scales/`); that sentence does not concern the prescribed-margin problem studied here. At that same raw-placement snapshot, no Lean source mentioned Sinkhorn scaling, spanning trees, doubly stochastic matrices or Kirchhoff’s theorem. These searches, like the original ones, locate words, not results stated under other names. The comparison with the two nearest reports is made precise in [section 1.5](#).

3 Hahn conventions and the summability issue

3.1 Fields, valuation, and positive leading coefficients

Throughout the local theory, Γ is a set-sized ordered abelian group, written additively. It need not be divisible, Archimedean, discrete, countable, or of finite rank. Put

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)). \quad (3.1)$$

A series $x = \sum_{\gamma \in S} x_\gamma t^\gamma$ has well-ordered support $S \subseteq \Gamma$. For $x \neq 0$ define $v(x) = \min S$ and $\text{lc}(x) = x_{v(x)}$; set $v(0) = +\infty$. The valuation ring and its maximal ideal are

$$\mathcal{O} = \{x : v(x) \geq 0\}, \quad \mathfrak{m} = \{x : v(x) > 0\}.$$

These symbols refer to K unless a subscript indicates F . A vector is infinitesimal when every coordinate belongs to $\mathfrak{m}$. All vector dimensions in this article are ordinary finite integers.

Definition 3.1. A nonzero $p \in K$ is *positive-leading* when $\text{lc}(p)$ is a positive real number. Equivalently, $p = at^\gamma(1 + u)$ with $a \in \mathbb{R}_{>0}$ and $u \in \mathfrak{m}$.

Every positive element of F is positive-leading. Finite nonempty sums and products, and inverses, preserve the positive-leading property. For a finite positive-leading family,

$$v\left(\sum_j p_j\right) = \min_j v(p_j). \quad (3.2)$$

Indeed, the coefficients at the least occurring exponent add to a positive real number. This is the only positivity mechanism required for the local surcomplex results. It does not put an order on K .

3.2 Strong summability

Definition 3.2. A set-indexed family $(x_i)_{i \in I}$ of Hahn series is *strongly summable* if $\bigcup_i \text{supp}(x_i)$ is well ordered and every exponent belongs to the support of only finitely many x_i . Its sum is formed coefficientwise, by finite sums at each exponent.

We will never infer strong summability solely from a statement that valuations increase strictly. Nor will a strongly summable expansion be called a limit of finite partial sums unless the relevant topological convergence has separately been justified.

Lemma 3.3 (Positive-support word finiteness). *If $S \subseteq \Gamma_{>0}$ is well ordered, then the additive monoid S^* generated by S , including 0, is well ordered. Every $\gamma \in \Gamma$ has only finitely many representations as a finite nondecreasing word of elements of S whose sum is γ . Consequently it has only finitely many ordered-word representations.*

Proof. Use Higman's theorem: finite words over a well-quasi-ordered alphabet are well-quasi-ordered by subsequence embedding with componentwise increase [5]. A well-ordered set is such an alphabet. If S^* had a strictly descending sequence, choose a nondecreasing word representing each of its terms. Some earlier word embeds into a later one. Positivity of all letters and nonnegativity of the componentwise increases imply that the earlier sum is at most the later sum, a contradiction.

If infinitely many distinct nondecreasing words had one fixed sum, apply the same theorem. An embedding can preserve the sum only if there are no extra letters and no strict componentwise increases. The two words would then be identical, again a contradiction. Each nondecreasing word has only finitely many permutations, proving the last assertion. $\square$

Lemma 3.4 (Finite-generator evaluation). *Let $a_1, \dots, a_s \in \mathcal{O}$ and $R_0 = \mathbb{C}[a_1, \dots, a_s] \subseteq \mathcal{O}$. For every formal series $\Phi(X) = \sum_{\nu \in \mathbb{N}^m} b_\nu X^\nu \in R_0[[X_1, \dots, X_m]]$ and every $x \in \mathfrak{m}^m$, the family $(b_\nu x^\nu)_\nu$ is strongly summable. Evaluation respects sums, products, and compositions with formal series of zero constant term whenever all coefficients lie in the same such finite-generator algebra.*

Proof. Let S be the union of the positive supports of the a_j and the supports of the nonzero x_i . It is a finite union of well-ordered positive sets, hence well ordered. Every a_j has only a possible constant term outside S . Since each b_ν is a polynomial in the a_j , the support of every evaluated term is contained in S^* .

More specifically, a contribution to $b_\nu x^\nu$ uses at least $|\nu|$ positive letters from the supports of the x_i . For a fixed exponent γ , lemma 3.3 bounds the length of all words summing to γ . Thus only bounded total degrees $|\nu|$ can contribute. There are only finitely many such multi-indices. This proves local finiteness at γ , while S^* proves well-ordering of the union of supports. If an input coordinate is zero, terms using a positive power of it vanish and can simply be omitted. For each of the finitely many relevant coefficients, its polynomial expression in the a_j has finitely many terms. Constant contributions of the generators cause no extra infinite sum. Thus no uniform bound on the polynomial degrees of all the b_ν is required.

For completeness, let $G_1, \dots, G_\ell \in R_0[[X]]$ have zero constant term and let $\Phi \in R_0[[Y_1, \dots, Y_\ell]]$. Each $G_j(x)$ lies in $\mathfrak{m}$: every nonzero evaluated term has positive input degree, and the union of its supports is well ordered and positive. At a fixed Hahn exponent, word finiteness bounds the total input degree. Since each inner monomial has positive degree, this also bounds the outer degree and leaves only finitely many choices of inner monomials. Expanding their finitely many coefficients therefore licenses the rearrangement

$$\text{ev}_x(\Phi \circ G) = \text{ev}_{G(x)}(\Phi).$$

The same finite-contribution argument proves compatibility with sums and products. This proves the composition assertion, including the case of zero inner series, without a topological limit. $\square$

Example 3.5 (Why integral coefficients alone do not suffice). In $\Gamma = \mathbb{Z}_{\text{lex}}^2$, let $\delta = (0, 1)$ and $\beta = (1, 0)$. The formal series

$$\Phi(X) = \sum_{n \geq 1} t^{\beta - n\delta} X^n$$

belongs to $\mathcal{O}[[X]]$, since $\beta - n\delta > 0$ for every n . But at $X = t^\delta$ every displayed term is t^β . The family is not strongly summable. The coefficients do not admit the common positive-support control in [lemma 3.4](#). An arbitrary-rank implicit-function proof that stops after establishing membership in $\mathcal{O}[[X]]$ therefore has a genuine gap.

3.3 Restricted exponential and logarithm

For $z \in \mathfrak{m}$ define

$$\exp z = \sum_{n \geq 0} \frac{z^n}{n!}, \quad \log(1 + z) = \sum_{n \geq 1} \frac{(-1)^{n+1} z^n}{n}. \quad (3.3)$$

The preceding lemma applies with constant coefficient algebra $\mathbb{C}$. The formal identities for these series remain valid under evaluation. In particular they are inverse group isomorphisms $\mathfrak{m} \leftrightarrow 1 + \mathfrak{m}$, and for nonzero infinitesimals

$$v(\exp z - 1) = v(z), \quad v(\log(1 + z)) = v(z). \quad (3.4)$$

Their leading terms are z . We also use

$$\varphi(z) = \frac{\exp z - 1}{z} = \sum_{n \geq 0} \frac{z^n}{(n+1)!}, \quad \varphi(0) = 1. \quad (3.5)$$

Thus $\varphi(z) \in 1 + \mathfrak{m}$. No exponential or logarithm at an arbitrary infinite surcomplex argument is used anywhere.

Remark 3.6 (The same support engine in the spectral report). [Lemma 3.3](#) is also Lemma 9.4 of the collection's spectral report [11], proved there from Higman's theorem as well, and [lemma 3.4](#) with no generators gives the strong-evaluation part of Lemma 9.5 there. The generators $a_1, \dots, a_s$ are what this report adds, and [example 3.5](#) shows that they cannot be replaced by arbitrary elements of $\mathcal{O}$. [Section 1.5](#) compares the two reports.

4 The tree interpolation identity

4.1 Incidence, cuts, and weighted cycles

Let $\mathcal{G} = (V, E)$ be a finite connected graph without loops, with an arbitrary orientation, $N = |V| \geq 2$ and $m = |E|$. Choose a root vertex. The reduced incidence matrix $B \in \mathbb{Z}^{(N-1) \times m}$ has $+1$ at the initial vertex and -1 at the terminal vertex of each edge, with the root row deleted. It has rank $N - 1$.

For positive-leading weights $p = (p_e)$ put

$$W_p = \text{diag}(p_e), \quad L_p = BW_pB^\top, \quad \Pi_p = B^\top L_p^{-1}BW_p, \quad H_p = I - \Pi_p. \quad (4.1)$$

The invertibility in this formula will follow from the tree identity. The subspace $\text{im } B^\top$ is the space of potential differences, or cuts. The kernel $\ker B$ is the space of conserved edge flows. The kernel $\ker(BW_p)$ consists of edge functions whose weighted flows are conserved.

For a spanning tree T , let B_T be the square submatrix with columns in T . Given $f \in K^E$, solve $B_T^\top u_T = f_T$ and define

$$\Pi_T f = B^\top u_T. \quad (4.2)$$

It extends the values of f on tree edges to a potential difference on every edge. Its value on an edge is the signed sum along the unique tree path between that edge's endpoints. Consequently every matrix entry of Π_T is 0, 1, or -1 .

Theorem 4.1 (Weighted tree interpolation). *Write*

$$w_T(p) = \prod_{a \in T} p_a, \quad Z(p) = \sum_{T \in \mathcal{T}} w_T(p).$$

Then

$$\det L_p = Z(p) \neq 0, \quad \Pi_p = \sum_{T \in \mathcal{T}} \frac{w_T(p)}{Z(p)} \Pi_T. \quad (4.3)$$

Moreover

$$\Pi_p^2 = \Pi_p, \quad H_p^2 = H_p, \quad H_p B^\top = 0, \quad BW_p H_p = 0, \quad (4.4)$$

and both Π_p and H_p have entries in $\mathcal{O}$.

Proof. Cauchy–Binet gives

$$\det(BW_p B^\top) = \sum_{|T|=N-1} (\det B_T)^2 \prod_{a \in T} p_a.$$

An incidence minor vanishes unless T is a spanning tree; for a tree it is ± 1 . A connected selection of $N - 1$ edges is a tree; deleting a leaf other than the root and expanding its row proves the ± 1 assertion by induction. A disconnected selection has deficient rank: a component not containing the root supplies a nonzero left-kernel vector, its vertex indicator. This proves the determinant formula. Every summand is positive-leading, so (3.2) proves nonvanishing.

To obtain the vector identity, set $u = L_p^{-1} BW_p f$. For coordinate k , replace column k of $B^\top$ by f , calling the resulting $m \times (N - 1)$ matrix C_k . Then $BW_p C_k$ is L_p with column k replaced by $BW_p f$. Cauchy–Binet and Cramer’s rule yield the following formula, where $(C_k)_T$ selects rows indexed by T , whereas B_T selects columns:

$$u_k = \frac{1}{Z(p)} \sum_T \det(B_T) w_T(p) \det((C_k)_T) = \frac{1}{Z(p)} \sum_T w_T(p) (u_T)_k.$$

In the second equality we used $\det((C_k)_T) = \det(B_T^\top) (u_T)_k$. Multiplying by $B^\top$ proves the interpolation formula.

The projection identities follow directly from $L_p = BW_p B^\top$. More explicitly, Π_p takes values in $\text{im } B^\top$ and fixes that space, since $\Pi_p B^\top = B^\top$. Also $BW_p \Pi_p = BW_p$, so $\Pi_p f = 0$ implies $BW_p f = 0$; the converse is immediate from its formula. Hence

$$\text{im } \Pi_p = \text{im } B^\top, \quad \ker \Pi_p = \ker(BW_p), \quad \text{im } H_p = \ker(BW_p), \quad \ker H_p = \text{im } B^\top.$$

These descriptions explain the complementary projections used below. Finally $v(w_T/Z) \geq 0$ by the determinant formula and no leading cancellation. Each Π_T has integer entries. Finite sums therefore show $\Pi_p, H_p \in \text{Mat}_m(\mathcal{O})$. $\square$

4.2 The row formula and the diagonal coefficient

For $e \notin T$, define the fundamental-cycle functional

$$\ell_{T,e}(f) = f_e - (\Pi_T f)_e.$$

It has coefficients 0, 1, -1 , its coefficient at f_e is 1, and it vanishes on $\text{im } B^\top$. If $e \in T$, the corresponding row of $I - \Pi_T$ is zero. Thus

$$(H_p f)_e = \sum_{\substack{T \in \mathcal{T} \\ e \notin T}} \frac{w_T(p)}{Z(p)} \ell_{T,e}(f). \quad (4.5)$$

In particular, with $Z_e^-(p) = \sum_{T:e \notin T} w_T(p)$,

$$(H_p)_{ee} = \frac{Z_e^-(p)}{Z(p)}. \quad (4.6)$$

For a bridge the numerator and the entire row of H_p vanish. For other edges its valuation is exactly the gap κ_e in (1.1).

The matrix L_p^{-1} itself need not be integral. The finite projection $B^\top L_p^{-1} B W_p$ nevertheless is integral. This cancellation of apparent conditioning loss is the algebraic reason to work with edge variables rather than estimate vertex potentials using an unstructured inverse bound.

Remark 4.2 (Positive tree sums elsewhere in the collection). That a positive-leading tree sum has the minimum tree cost as its valuation, and the sum of the minimizers' leading amplitudes as its leading coefficient, is the mechanism of the directed forest sums of the Markov report [10] and of the squared-minor sums of Gram determinants in the spectral report [11]. Section 1.5 states what is shared and what is not.

5 Normalization on the whole infinitesimal polydisc

5.1 The nonlinear equations

Keep the finite oriented graph and positive-leading weights from the previous section. For $f \in \mathfrak{m}^E$ we seek $h \in \mathfrak{m}^E$ satisfying

$$h - f \in \text{im } B^\top, \quad B W_p(\exp h - \mathbf{1}) = 0. \quad (5.1)$$

Exponential, multiplication, and subtraction in an edge vector are componentwise. The corresponding normalized weights are

$$q_p(f) = p \odot \exp h. \quad (5.2)$$

The second equation says $B q_p(f) = B p$. On a bipartite graph oriented from rows to columns it says exactly that every row sum and column sum is restored. Deleting one incidence row loses no condition, because the sum of all rows of the full incidence matrix is zero. The first equation says that $q_p(f)$ is obtained from $p \odot \exp f$ by vertex-potential factors.

Theorem 5.1 (Uniform infinitesimal normalization). *For every $f \in \mathfrak{m}^E$ there is exactly one $h \in \mathfrak{m}^E$ satisfying (5.1). Write $h = \mathcal{N}_p(f)$. There is a unique formal series*

$$\mathcal{H}_p(X) \in (X_1, \dots, X_m) \mathcal{O}[[X_1, \dots, X_m]]^E$$

solving the same equations formally. Its coefficients belong to the finite-generator algebra

$$R_p = \mathbb{C}[(\Pi_p)_{ab} : a, b \in E] \subseteq \mathcal{O}. \quad (5.3)$$

It evaluates strongly at every $f \in \mathfrak{m}^E$, and that evaluation is $\mathcal{N}_p(f)$. No lower bound on the entries p_e , no rank-one hypothesis, and no divisibility of Γ are required.

Proof. Put $R(z) = \exp z - 1 - z$, componentwise. Applying the complementary projections to (5.1) shows that these equations are equivalent to

$$h = H_p f - \Pi_p R(h). \quad (5.4)$$

Indeed, the cut condition is $H_p h = H_p f$, and conservation is $\Pi_p(h + R(h)) = 0$. Adding these two projected identities gives (5.4); the converse follows from theorem 4.1. The image

and kernel descriptions extend coefficientwise to the modules over $K[[X]]$, giving the same equivalence for the formal equations.

Now work formally in variables $X = (X_e)$. Write $\mathcal{H}_p = \sum_{d \geq 1} h_d$ with each h_d homogeneous of degree d . Since R starts in degree two, the recurrence is triangular:

$$h_1 = H_p X, \quad (5.5)$$

$$h_d = -\Pi_p \sum_{k=2}^d \frac{1}{k!} \sum_{\substack{d_1 + \dots + d_k = d \\ d_j \geq 1}} h_{d_1} \odot \dots \odot h_{d_k}, \quad d \geq 2. \quad (5.6)$$

Every index d_j on the right is less than d . This constructs one and only one formal solution. Induction shows that its coefficients are polynomials over $\mathbb{Q}$ in the entries of Π_p ; $H_p = I - \Pi_p$ requires no extra generators. Thus they belong to $R_p \subseteq \mathcal{O}$.

Apply lemma 3.4 to this specific coefficient algebra. It proves strong evaluation of the entire series at $f \in \mathfrak{m}^E$, not merely formal existence. It also permits the products and exponential substitution in (5.4). The evaluated vector lies in $\mathfrak{m}^E$: each term has positive degree in infinitesimal inputs, and its support lies in a well-ordered positive set. Hence it gives a solution of (5.1).

For uniqueness, let $h, k \in \mathfrak{m}^E$ solve (5.4). Suppose $h \neq k$ and let $\nu = \min_e v(h_e - k_e)$. There is a common $\eta > 0$ not exceeding the valuations of any nonzero coordinate of h or k , since there are finitely many. The formal factorization

$$R(z) - R(w) = (z - w)\psi(z, w), \quad \psi(z, w) = \sum_{n \geq 2} \frac{1}{n!} \sum_{j=0}^{n-1} z^{n-1-j} w^j \in (z, w)\mathbb{C}[[z, w]],$$

together with strong evaluation gives

$$\min_e v(R(h_e) - R(k_e)) \geq \nu + \eta.$$

But $h - k = -\Pi_p(R(h) - R(k))$, and every entry of Π_p is integral. It follows that $\nu \geq \nu + \eta$, impossible in an ordered group. This argument uses one strict valuation improvement, not convergence of a sequence of improvements. It is therefore valid at arbitrary rank. $\square$

5.2 Gauge factors, retraction, and local coordinates

Corollary 5.2. *There is a unique root-normalized potential $u \in \mathfrak{m}^{N-1}$ with $\mathcal{N}_p(f) - f = B^\top u$. Thus in the matrix case the normalizing row factors and column factors may all be chosen in $1 + \mathfrak{m}$. Furthermore, for $a \in \mathfrak{m}^{N-1}$,*

$$\mathcal{N}_p(f + B^\top a) = \mathcal{N}_p(f). \quad (5.7)$$

The map is a retraction onto the set of $h \in \mathfrak{m}^E$ satisfying $BW_p(\exp h - \mathbf{1}) = 0$.

Proof. Existence of a potential over K follows from the cut condition. Restrict that condition to any spanning tree. The inverse of $B_T^\top$ has integer entries 0, 1, -1 , given by signed root-path sums. Thus the root-normalized potential is a finite signed sum of infinitesimal edge differences and is itself infinitesimal. Uniqueness follows from the rank of B . For an edge oriented from row i to column j , the multiplier is $\exp(u_i - u_j)$, with the root potential interpreted as zero. These are row factors $\exp u_i$ and column factors $\exp(-u_j)$. Changing f by a cut does not change its class modulo cuts, so uniqueness proves (5.7). If h is already balanced, it solves the problem with input h , proving the retraction assertion. $\square$

The weighted cycle coordinates are especially simple:

$$H_p \mathcal{N}_p(f) = H_p f. \quad (5.8)$$

Consequently $x \mapsto \mathcal{N}_p(x)$ is a bijection from $\ker(BW_p) \cap \mathfrak{m}^E$ onto the balanced infinitesimal set, with inverse $h \mapsto H_p h$. Indeed $H_p \mathcal{N}_p(x) = H_p x = x$ on that kernel. Conversely, for balanced h , gauge invariance and retraction give $\mathcal{N}_p(H_p h) = \mathcal{N}_p(h) = h$; $H_p h - h$ is an infinitesimal cut. Integrality of H_p ensures that these maps preserve the infinitesimal domains. At a balanced point h , the formal linearization of conservation has kernel $\ker(B \operatorname{diag}(p_e \exp h_e))$. The diagonal is invertible and B has rank $N - 1$, so this formal tangent space has dimension $m - N + 1$ over K ; at $h = 0$ it is $\ker(BW_p)$. This is a support-controlled analytic chart in Hahn coordinates, not an assertion about classical holomorphic manifolds in the full surreal fine topology.

For reference, the first terms are

$$\begin{aligned} h_1 &= H_p X, \\ h_2 &= -\frac{1}{2} \Pi_p(h_1 \odot h_1), \\ h_3 &= \frac{1}{2} \Pi_p(h_1 \odot \Pi_p(h_1 \odot h_1)) - \frac{1}{6} \Pi_p(h_1 \odot h_1 \odot h_1). \end{aligned} \quad (5.9)$$

Only the first line is the familiar implicit differentiation formula. The theorem supplies all orders and justifies their evaluation on a common relative infinitesimal domain.

6 Sharp nonlinear valuation gains

6.1 Secant reweighting

The decisive step turns an exact nonlinear comparison into a linear projection with slightly altered weights.

Lemma 6.1 (Exact secant identity). *Let $f, g \in \mathfrak{m}^E$, set $h = \mathcal{N}_p(f)$, $k = \mathcal{N}_p(g)$, $r = h - k$, and $d = f - g$. Define*

$$\tilde{p}_e = p_e \exp(k_e) \varphi(r_e). \quad (6.1)$$

Then $\tilde{p}_e$ has the same valuation and leading coefficient as p_e , and

$$r = H_{\tilde{p}} d. \quad (6.2)$$

Proof. Both $\exp(k_e)$ and $\varphi(r_e)$ belong to $1 + \mathfrak{m}$, so the leading data are unchanged. Subtract the two conservation equations:

$$0 = B \operatorname{diag}(p_e \exp k_e)(\exp r - \mathbf{1}) = B \operatorname{diag}(\tilde{p}_e) r.$$

Also $r - d \in \operatorname{im} B^\top$. Project with $H_{\tilde{p}}$: it fixes r by the first identity and kills $r - d$ by the second. This gives (6.2). $\square$

6.2 The optimal two-point bound

For a tree T put $c(T) = \sum_{a \in T} c_a$, with $c_a = v(p_a)$. Define τ , τ_e^- , and κ_e by (1.1). If e is not a bridge, let

$$C_e = \frac{\sum_{T: e \notin T} \prod_{a \in T} \operatorname{lc}(p_a)}{\sum_{\substack{T \\ c(T) = \tau}} \prod_{a \in T} \operatorname{lc}(p_a)} \in \mathbb{R}_{>0}. \quad (6.3)$$

Ties are included in both sums.

Theorem 6.2 (Sharp nonlinear tree-gap law). *For every pair $f, g \in \mathbf{m}^E$ with $f \neq g$, put $\delta = \min_a v(f_a - g_a) > 0$. Then for every edge e ,*

$$v(\mathcal{N}_p(f)_e - \mathcal{N}_p(g)_e) = v\left(\frac{q_p(f)_e}{q_p(g)_e} - 1\right) \geq \delta + \kappa_e. \quad (6.4)$$

If e is a bridge, its normalized entry is exactly p_e for all inputs. If e is not a bridge and $f - g = s\mathbf{e}_e$ with $0 \neq s \in \mathbf{m}$, then

$$v(\mathcal{N}_p(f)_e - \mathcal{N}_p(g)_e) = v(s) + \kappa_e, \quad (6.5)$$

and, more precisely,

$$\text{lc}\left(\frac{\mathcal{N}_p(f)_e - \mathcal{N}_p(g)_e}{s}\right) = C_e, \quad v\left(\frac{\mathcal{N}_p(f)_e - \mathcal{N}_p(g)_e}{s}\right) = \kappa_e. \quad (6.6)$$

The same leading amplitude holds with the logarithmic difference replaced by $q_p(f)_e/q_p(g)_e - 1$.

Proof. Apply the row formula (4.5) to the exact secant identity:

$$r_e = \sum_{T: e \notin T} \frac{w_T(\tilde{p})}{Z(\tilde{p})} \ell_{T,e}(d). \quad (6.7)$$

Every cycle functional has valuation at least δ , since its coefficients are $0, 1, -1$. All reweighted entries have their original leading data, so no leading cancellation occurs in the tree sums. In particular

$$vZ(\tilde{p}) = \tau, \quad v(w_T(\tilde{p})/Z(\tilde{p})) = c(T) - \tau.$$

Each tree appearing in (6.7) avoids e , hence each summand has valuation at least $\delta + \kappa_e$. This proves the inequality. For a bridge the sum is empty; taking $g = 0$ gives $r_e = 0$ and $q_p(f)_e = p_e$.

If $d = s\mathbf{e}_e$, then $\ell_{T,e}(d) = s$ for every tree avoiding e . The exact identity becomes

$$r_e = s \frac{Z_e^-(\tilde{p})}{Z(\tilde{p})}. \quad (6.8)$$

Both finite sums are nonzero and positive-leading. Their valuations are τ_e^- and τ , and their leading coefficients are the numerator and denominator in (6.3). This proves (6.5) and (6.6), without a genericity assumption. Finally $q_p(f)_e/q_p(g)_e = \exp r_e$, so (3.4) gives equality of valuations and preservation of the leading coefficient for the relative output error. $\square$

Remark 6.3 (What “sharp” means). If $\Gamma \neq \{0\}$, the quantity κ_e is the largest gain valid uniformly over all infinitesimal changes and all centers in this relative polydisc: any nonzero $s \in \mathbf{m}$ tests its own coordinate. For $\Gamma = \{0\}$ there is only the zero input, so this evaluated optimality assertion is replaced by the formal coefficient assertion below. A particular perturbation can have a larger gain because of cycle cancellation or because it is supported far away. The chain theorem below gives exact examples of such stronger, location-sensitive gains. Sharpness does not assert equality for every input direction.

6.3 All Taylor orders and a remainder certificate

Theorem 6.4 (Coefficientwise gain). *For each edge e and each nonzero multi-index ν ,*

$$v([X^\nu]\mathcal{H}_p(X)_e) \geq \kappa_e, \quad v([X^\nu](\exp \mathcal{H}_p(X)_e - 1)) \geq \kappa_e. \quad (6.9)$$

For a nonbridge, the coefficient of X_e has valuation exactly κ_e and leading coefficient C_e . At every center $g \in \mathfrak{m}^E$ the same assertions hold for the centered branch with base weights $q_p(g)$.

Proof. Set formally $\tilde{p}_e = p_e \varphi(\mathcal{H}_p(X)_e)$. The formal weighted Laplacian has constant term L_p , so its determinant has nonzero constant term $Z(p)$ and is a unit in $K[[X]]$. The adjugate, Cauchy–Binet and Cramer identities used for tree interpolation therefore apply over this formal ring as well. No notion of positive-leading X -series is needed. The formal secant argument gives

$$\mathcal{H}_p(X)_e = \sum_{T: e \notin T} \frac{w_T(\tilde{p})}{Z(\tilde{p})} \ell_{T,e}(X).$$

The denominator can be written as

$$\frac{Z(\tilde{p})}{Z(p)} = \sum_T \frac{w_T(p)}{Z(p)} \prod_{a \in T} \varphi(\mathcal{H}_p(X)_a).$$

It has coefficients in $\mathcal{O}$ and constant term 1. Therefore its formal inverse has coefficients in $\mathcal{O}$. For every tree avoiding e , the scalar $w_T(p)/Z(p)$ has valuation at least κ_e . All other factors in the row formula have integral coefficients. The first bound in (6.9) follows. For a bridge the row sum is empty, so the entire formal row is zero and both bounds with $\kappa_e = +\infty$ hold. For a nonbridge, since $\kappa_e \geq 0$, every positive power of that row series also has all its nonzero coefficients of valuation at least κ_e ; this proves the exponential bound. The coefficient of X_e is $(H_p)_{ee} = Z_e^-(p)/Z(p)$, with the asserted leading data.

For a center g , put $h = \mathcal{N}_p(g + x)$, $k = \mathcal{N}_p(g)$ and $r = h - k$. Then $r - x \in \text{im } B^\top$ and

$$B \text{diag}(q_p(g))(\exp r - \mathbf{1}) = B \text{diag}(p)(\exp h - \exp k) = 0.$$

The vector r is infinitesimal, and $q_p(g)$ has the same positive leading data as p . Uniqueness at this new base gives the exact recentering identity

$$\mathcal{N}_p(g + x) - \mathcal{N}_p(g) = \mathcal{N}_{q_p(g)}(x), \quad x \in \mathfrak{m}^E. \quad (6.10)$$

Its right side is defined by its own support-controlled formal series, so no unjustified substitution of a nonzero constant into an arbitrary formal series is involved. The weights $q_p(g)$ have the same valuations and leading coefficients as p . Apply the proved result with that base. $\square$

Corollary 6.5 (Finite-degree precision certificate). *If $D \in \mathbb{N}_0$, $f \in \mathfrak{m}^E$ is nonzero, $\delta = \min_e v(f_e)$, and $h^{[D]} = \sum_{d=1}^D h_d(f)$, then*

$$v(\mathcal{N}_p(f)_e - h_e^{[D]}) \geq \kappa_e + (D + 1)\delta. \quad (6.11)$$

An analogous bound holds for truncating the relative output series.

Proof. Every term of total degree $d > D$ has valuation at least $\kappa_e + d\delta$. The family is strongly summable by the existence theorem. Its sum cannot acquire an exponent below the common lower bound $\kappa_e + (D + 1)\delta$. The relative output series $\exp(\mathcal{H}_p) - \mathbf{1}$ also has coefficients in R_p and zero constant term, so the same evaluation lemma and coefficient bound apply. The case $D = 0$ uses the empty truncation; bridge rows are identically zero. $\square$

This is an exact certificate, but it is not a promise that finite Taylor degree reaches every requested precision. In a higher-rank group the set of multiples $D\delta$ can be bounded above. The complete Hahn value still exists by its support construction.

7 Computing the gain without summing trees

Theorem 7.1 (Bottleneck replacement formula). *For a nonbridge edge e joining vertices u and v , define*

$$b_e = \min_{\pi: u \rightsquigarrow v \text{ in } \mathcal{G} \setminus e} \max_{a \in \pi} c_a,$$

where it suffices to consider simple paths. Then

$$\kappa_e = \max(0, b_e - c_e). \quad (7.1)$$

Thus $\kappa_e > 0$ precisely when e belongs to every minimum-valuation spanning tree; $\kappa_e = 0$ precisely when some minimum tree avoids e .

Proof. The finite nonempty set of simple alternative paths attains its minimum. Choose a minimum-weight spanning tree T_0 of $\mathcal{G} \setminus e$. The maximum edge cost on its u - v path is b_e . To prove this familiar bottleneck property, remove a maximum-cost edge of that path. Every alternative u - v path crosses the resulting cut. If an alternative path had every cost below that maximum, one of its crossing edges could replace the removed edge and make T_0 cheaper, a contradiction.

Insert e into T_0 and remove a path edge of cost b_e . The resulting tree has total cost $\tau_e^- + c_e - b_e$. On the other hand, let U be any tree containing e . Removing e separates it into two components. A path realizing b_e crosses this cut in some edge a of cost at most b_e . Hence

$$\tau_e^- \leq c(U) - c_e + c_a \leq c(U) - c_e + b_e.$$

It follows that the least tree cost in the full graph is

$$\tau = \min\{\tau_e^-, \tau_e^- + c_e - b_e\} = \tau_e^- + \min(0, c_e - b_e).$$

Subtracting from τ_e^- proves (7.1). The last assertions also follow directly from the definitions of τ and τ_e^- . $\square$

All calculations in this proof use finite sums, comparisons, and subtractions in Γ . No real embedding of the value group is needed. A simple exact certificate for all gains consists of one minimum spanning tree and replacement-edge data for its cuts. An edge outside the chosen minimum tree already has gain zero. For a tree edge, deleting it gives a cut; the cheapest other crossing edge determines its replacement cost. The difference from the original edge cost is its gain, or $+\infty$ when no replacement exists. To justify this certificate for any chosen minimum tree T , let f be the cheapest replacement for $e \in T$. For any tree U avoiding e , its endpoint path crosses the cut of $T \setminus e$ in an edge a with $c_a \geq c_f$. The tree $U + e - a$ and minimality of T give

$$c(T) \leq c(U) + c_e - c_a \leq c(U) + c_e - c_f.$$

Thus $T - e + f$ attains $\tau_e^- = c(T) - c_e + c_f$. This argument includes ties. These are standard finite minimum-spanning-tree operations; the new use is the precision certificate for a nonlinear Hahn normalization.

The positive leading coefficient C_e contains more information than κ_e . Computing it requires the weighted sums of all minimizing trees in (6.3), or equivalent determinant initial terms. A single minimum tree does not determine that amplitude when there are ties.

8 Exact nonlinear propagation through a chain

8.1 The matrix and the response

Let $n \geq 2$ and choose positive infinitesimals $a_1, \dots, a_{n-1} \in F$, with $\gamma_i = v(a_i) > 0$. Put $a_0 = a_n = 0$ and

$$p_{ii} = d_i = 1 - a_{i-1} - a_i, \quad p_{i,i+1} = p_{i+1,i} = a_i, \quad (8.1)$$

with all other entries zero. The matrix P is positive on its tridiagonal support and is bistochastic. Each d_i has valuation zero and leading coefficient 1.

Perturb only the bottom-right kernel entry:

$$A_{nn} = p_{nn} \exp s, \quad A_{ij} = p_{ij} \quad ((i, j) \neq (n, n)), \quad 0 \neq s \in \mathfrak{m}_K.$$

Let Q be the normalization from [theorem 5.1](#), and write $\delta = v(s)$. The result below gives all nonzero response valuations, not just a common bound.

Theorem 8.1 (Additive attenuation along the chain). *For $1 \leq i \leq n-1$,*

$$v\left(\frac{Q_{i,i+1}}{a_i} - 1\right) = v\left(\frac{Q_{i+1,i}}{a_i} - 1\right) = \delta + \sum_{k=i+1}^{n-1} \gamma_k. \quad (8.2)$$

For the diagonal entries,

$$v\left(\frac{Q_{ii}}{d_i} - 1\right) = \delta + \sum_{k=i}^{n-1} \gamma_k, \quad 1 \leq i < n, \quad (8.3)$$

$$v\left(\frac{Q_{nn}}{d_n} - 1\right) = \delta + \gamma_{n-1}. \quad (8.4)$$

These are exact nonlinear identities for arbitrary positive γ_i in the ordered group. They do not require the γ_i to be comparable by Archimedean size.

8.2 Cycle coordinates and the secant matrix

Proof. Let $r = n - 1$. For $1 \leq i \leq r$, define the edge vector C_i to have $+1$ at (i, i) and $(i+1, i+1)$, -1 at $(i, i+1)$ and $(i+1, i)$, and zero elsewhere. These are conserved flows. They are linearly independent because each off-diagonal pair belongs to only one C_i . The connected support graph has $2n$ vertices and $3n - 2$ edges, so its cycle-space dimension is $n - 1$. Hence the columns of $C = (C_1, \dots, C_r)$ form a basis of $\ker B$.

Write the change in normalized entries as

$$j = Q - P = Cx. \quad (8.5)$$

Then

$$j_{i,i+1} = j_{i+1,i} = -x_i, \quad j_{ii} = x_{i-1} + x_i,$$

where $x_0 = x_n = 0$. Set $h = \mathcal{N}_p(\mathbf{se}_{nn})$ and $\tilde{p}_e = p_e \varphi(h_e)$. The identity $Q - P = \text{diag}(\tilde{p}_e)h$ gives $h = \text{diag}(\tilde{p}_e)^{-1}Cx$. Since $C^\top B^\top = 0$, the cut condition implies $C^\top h = C^\top(\mathbf{se}_{nn}) = \mathbf{se}_r$. Thus

$$Mx = \mathbf{se}_r, \quad M = C^\top \text{diag}(\tilde{p}_e)^{-1}C. \quad (8.6)$$

This equation is exact even though its coefficients depend on the unknown normalized solution. That dependence is harmless for valuations, because the leading data of $\tilde{p}$ and p coincide.

The matrix M is symmetric tridiagonal. Its entries are

$$\begin{aligned} M_{ii} &= \widetilde{p}_{ii}^{-1} + \widetilde{p}_{i+1,i+1}^{-1} + \widetilde{p}_{i,i+1}^{-1} + \widetilde{p}_{i+1,i}^{-1}, \\ M_{i,i+1} &= M_{i+1,i} = \widetilde{p}_{i+1,i+1}^{-1}. \end{aligned}$$

Consequently

$$v(M_{ii}) = -\gamma_i, \quad v(M_{i,i+1}) = 0, \quad M_{ii} \sim \frac{2}{a_i}, \quad M_{i,i+1} \sim 1. \quad (8.7)$$

Here $u \sim v$ means $u/v \in 1 + \mathfrak{m}$. The two inverse off-diagonal weights in M_{ii} have the same positive leading coefficient and cannot cancel.

Let $D_{a,b}$ be the determinant of the principal interval indexed from a to b , with the empty determinant equal to 1. The initial values are $D_{a,a} = M_{aa}$ and the empty determinant 1. For $b \geq a + 1$ the continuant recurrence is

$$D_{a,b} = M_{bb}D_{a,b-1} - M_{b-1,b}^2 D_{a,b-2}.$$

Induction proves

$$v(D_{a,b}) = -\sum_{k=a}^b \gamma_k, \quad D_{a,b} \sim \prod_{k=a}^b \frac{2}{a_k}. \quad (8.8)$$

For an interval of length at least two, the second term in the recurrence has valuation strictly larger than that of the first by $\gamma_{b-1} + \gamma_b > 0$. This proves both statements and also proves that M is invertible.

For $i \leq j$, cofactor expansion of a tridiagonal matrix gives

$$(M^{-1})_{ij} = (-1)^{i+j} \frac{\left(\prod_{k=i}^{j-1} M_{k,k+1}\right) D_{1,i-1} D_{j+1,r}}{D_{1,r}}. \quad (8.9)$$

One way to see the factorization is to delete row j and column i in the cofactor. Expanding successively through the intervening rows forces the entries $M_{i,i+1}, \dots, M_{j-1,j}$; the remaining blocks are the two displayed principal intervals. The cofactor sign is $(-1)^{i+j}$. Symmetry gives the formula for $i > j$. Equations (8.7)–(8.9) now imply

$$v((M^{-1})_{ij}) = \sum_{k=\min(i,j)}^{\max(i,j)} \gamma_k. \quad (8.10)$$

In particular,

$$v(x_i) = \delta + \sum_{k=i}^r \gamma_k, \quad x_i \sim (-1)^{r-i} \frac{s a_i a_{i+1} \cdots a_r}{2^{r-i+1}}. \quad (8.11)$$

These are exact statements about the nonlinear solution of (8.6), not substitutions into a linearization at $s = 0$.

For an off-diagonal entry, divide $-x_i$ by a_i to obtain (8.2). If $1 < i < n$, then $v(x_{i-1}) = v(x_i) + \gamma_{i-1} > v(x_i)$, so the two summands in $j_{ii} = x_{i-1} + x_i$ cannot cancel at the leading exponent. The first diagonal entry is $j_{11} = x_1$ and the last is $j_{nn} = x_r$. Since all d_i have valuation zero, the diagonal formulas follow. For the smallest chain $n = 2$, M has one entry, $x_1 \sim s a_1/2$, both diagonal relative responses have valuation $\delta + \gamma_1$, and both off-diagonal relative responses have valuation δ . This also checks the empty sums in the statement. $\square$

8.3 A rank-two example

Take $n = 3$, $a_1 = t^\alpha$, $a_2 = t^\beta$, with $\alpha = (0, 1)$ and $\beta = (1, 0)$ in $\mathbb{Z}_{\text{lex}}^2$. Let $s \neq 0$ be any surcomplex infinitesimal. The leading responses are

Entry	Relative error, up to a factor in $1 + \mathfrak{m}$	Valuation
$(1, 1)$	$-s t^{\alpha+\beta}/4$	$v(s) + \alpha + \beta$
$(2, 2), (3, 3)$	$s t^\beta/2$	$v(s) + \beta$
$(1, 2), (2, 1)$	$s t^\beta/4$	$v(s) + \beta$
$(2, 3), (3, 2)$	$-s/2$	$v(s)$

The perturbation reaches the adjacent small link at its original relative scale, but its influence on the far end acquires both link valuations. The formulas remain valid when $v(s) = \alpha$; there is no need for integer multiples of α to cross β . The far diagonal has uniform own-entry gain α , but this remote perturbation gains $\alpha + \beta$. The chain theorem is therefore strictly more location-sensitive than the uniform row bound. The signs in the table also illustrate why off-coordinate leading responses can alternate even though the sharp own-coordinate amplitude in [theorem 6.2](#) is positive.

9 Global positive matrix scaling and its relation to the branch

9.1 Classical existence and real-closed transfer

The following background proposition explains how the local result applies to the usual positive normalization, not only to an abstract choice of a surcomplex branch.

Proposition 9.1 (Positive scaling over real closed fields). *Let F_0 be a real closed ordered field. Fix a finite bipartite support E and prescribed row and column margins. Suppose a matrix P with exactly this support, strictly positive on E , realizes those margins. Then every matrix A strictly positive on the same support has a unique normalized matrix $Q = D_r A D_c$ with those margins, where D_r, D_c are positive diagonal matrices. On a connected support the factors are unique up to multiplying D_r by a common positive scalar and D_c by its inverse.*

Proof. Over $\mathbb{R}$, consider the nonnegative matrices on E with the given margins and minimize

$$\sum_{e \in E} x_e (\log(x_e/A_e) - 1), \quad 0 \log 0 := 0.$$

The feasible polytope is nonempty and compact: each entry is bounded by its row sum. The objective is strictly convex, so a minimizer exists and is unique. It has no zero entry on E . Indeed, interpolate from a putative boundary minimizer toward the strictly positive feasible witness P . A zero coordinate contributes a negative term of order $-c s |\log s|$, while changes in positive coordinates are only $O(s)$. The objective would decrease for sufficiently small positive real s .

At the interior minimizer, the gradient vector $(\log(x_e/A_e))$ annihilates the tangent space of the margin constraints. It is therefore a sum of a row potential and a column potential. Exponentiating gives the diagonal factors. Conversely, such a positive matrix is a constrained critical point and is the unique minimizer. If two pairs of factors give the same matrix, their ratios are reciprocal across each support edge; connectedness makes them one common gauge factor. This is a standard form of classical scaling theory [\[6, 7\]](#).

For each fixed finite support, the existence and uniqueness assertion can be written using only polynomial equalities, strict inequalities, and finitely many quantified field elements:

the positive witness P , the input entries, the margins, the factors, and the output entries. Completeness of the theory of real closed fields [8] transfers that assertion from $\mathbb{R}$ to F_0 . The real optimization proof is not repeated inside a non-Archimedean field, and no non-Archimedean compactness or global logarithm is assumed. $\square$

In particular, the proposition applies to $F = \mathbb{R}((t^\Gamma))$ when Γ is divisible. The local results earlier require no such hypothesis.

9.2 An elementary multiplicative comparison

Proposition 9.2 (A finite cut bound). *Over any ordered field, suppose two positive kernels A, A' on a connected bipartite graph with N vertices have positive diagonal normalizations P, P' to the same margins. If $\eta \geq 1$ and*

$$\eta^{-1} \leq A'_e/A_e \leq \eta \quad (e \in E),$$

then

$$\eta^{-N} \leq P'_e/P_e \leq \eta^N \quad (e \in E). \quad (9.1)$$

This is a useful elementary comparison bound; optimality of the ordinary exponent N is not claimed.

Proof. Write $P = D_r A D_c$ and $P' = D'_r A' D'_c$. Associate a positive vertex number $x_i = D'_{r,i}/D_{r,i}$ to a row vertex and $x_j = D_{c,j}/D'_{c,j}$ to a column vertex. Then

$$\frac{P'_{ij}}{P_{ij}} = \frac{A'_{ij}}{A_{ij}} \frac{x_i}{x_j}.$$

Sort the N vertex numbers in nondecreasing order. No consecutive ratio can exceed η . Otherwise, let U be the nonempty proper set of vertices above such a gap. On every edge from a row in U to a column outside U , one has $P'_e - P_e > 0$. On every edge from a row outside U to a column in U , one has $P'_e - P_e < 0$. With incidence signs, every crossing edge therefore contributes strictly positively to the total divergence over U . There is a crossing edge by connectedness. But P and P' have the same margins, so that divergence is zero, a contradiction.

The ratio of the largest vertex number to the smallest is thus at most η^{N-1} . Multiplying this bound by the kernel ratio bound proves (9.1) and its reciprocal. $\square$

Corollary 9.3 (The positive normalization is the local branch). *Assume Γ is divisible and A, A' are positive matrices over F on the same connected support, with fixed margins realized by a matrix strictly positive on every edge of that support. Let P, P' be their positive normalizations. If $A'_e/A_e \in 1 + \mathfrak{m}_F$ for every edge, put $f_e = \log(A'_e/A_e)$. Then*

$$P' = q_P(f).$$

In particular theorem 6.2 applies to the usual positive normalization, with the tree costs computed from the normalized base matrix P .

Proof. A finite maximum of the kernel ratios, their reciprocals, and 1 gives an $\eta \in 1 + \mathfrak{m}_F$ as in the cut bound. Its positive and negative integer powers also lie in $1 + \mathfrak{m}_F$. Since the infinitesimals are order-convex, the inequalities $\eta^{-N} \leq P'_e/P_e \leq \eta^N$ imply $P'_e/P_e \in 1 + \mathfrak{m}_F$. The vertex numbers used in that proof may be divided by a root value. Their largest-to-smallest

ratio is bounded by η^{N-1} , so every root-normalized ratio lies between $\eta^{-(N-1)}$ and η^{N-1} and thus in $1 + \mathfrak{m}_F$. Their restricted logarithms give an infinitesimal potential u with

$$\log(P'/P) - f = B^\top u.$$

The margin equation is automatic. Uniqueness in [theorem 5.1](#) now identifies P' with $q_P(f)$. $\square$

The support-feasibility hypothesis means more than positive margin numbers and a non-negative feasible matrix. On the support of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, row and column margins $(1, 1)$ force the off-diagonal entry to be zero; no positive diagonal scaling can realize them on the entire support.

The distinction between A and its normalized matrix P matters: the costs in the sharp theorem are valuations of the base *normalized entries*, not, in general, valuations of the unscaled kernel. We do not claim an equally sharp formula from unscaled valuations alone.

10 Extension to real linear constraints

The nonlinear support argument and sharp gain do not fundamentally require an incidence matrix. This gives a broader version useful for finite exponential constraint systems.

Theorem 10.1 (Basis-gap normalization). *Let $B \in \mathbb{R}^{r \times m}$ have full row rank and let $p \in K^m$ have positive-leading coordinates. Consider*

$$h - f \in \text{im } B^\top, \quad B \text{diag}(p_e)(\exp h - \mathbf{1}) = 0, \quad f, h \in \mathfrak{m}^m.$$

There is a unique solution for every f , represented by a support-controlled formal series on the whole infinitesimal polydisc. Let $\mathcal{B}$ be the column bases of B and define

$$\begin{aligned} \tau_B &= \min_{T \in \mathcal{B}} \sum_{a \in T} v(p_a), \\ \tau_{B,e}^- &= \min_{\substack{T \in \mathcal{B} \\ e \notin T}} \sum_{a \in T} v(p_a), \quad \kappa_{B,e} = \tau_{B,e}^- - \tau_B. \end{aligned}$$

An empty deletion minimum is $+\infty$. The nonlinear two-point inequality, one-coordinate equality, and all Taylor-coefficient bounds of [theorems 6.2](#) and [6.4](#) hold with these basis gaps. A column contained in every basis has an exactly fixed normalized coordinate. In the leading amplitude formula, each basis term acquires the factor $(\det B_T)^2$.

Proof. For a basis T put $\Pi_T f = B^\top (B_T^\top)^{-1} f_T$. Its coefficients are real constants and hence integral in K . Cauchy–Binet and the Cramer’s-rule argument give

$$Z_B(p) = \det(B \text{diag}(p) B^\top) = \sum_{T \in \mathcal{B}} (\det B_T)^2 \prod_{a \in T} p_a,$$

and

$$\Pi_p = \sum_{T \in \mathcal{B}} \frac{(\det B_T)^2 \prod_{a \in T} p_a}{Z_B(p)} \Pi_T.$$

Every determinant square in this sum is a strictly positive real constant. The denominator is positive-leading, the scalar ratios are integral, and Π_p, H_p have integral entries. The recurrence and finite-generator proof in [theorem 5.1](#) therefore apply unchanged.

For a coordinate $e \in T$, the e th row of $I - \Pi_T$ is zero. For $e \notin T$, the e th row is a real linear functional with coefficient 1 at f_e . Its value on an infinitesimal vector of minimum valuation

δ has valuation at least δ . The secant identity is purely linear algebra plus the exponential identity and remains valid. Repeating its rowwise proof gives the basis-gap bound and exact equality for $f - g = se_e$. The same formal denominator argument proves the coefficient bounds. The leading determinant-square factors are exactly those displayed in Z_B . If e belongs to every basis (a coloop), every row in the formula for $(H_p f)_e$ vanishes, so its normalized coordinate is exactly p_e . A zero column belongs to no basis and has gap zero; its row in Π_p is zero and its logarithmic response is precisely f_e . The convention also covers $r = 0$: the unique basis is empty, its determinant and weight are 1, and the normalization is the identity. If $m = 0$, full row rank forces $r = 0$ and there are no coordinate assertions. $\square$

The hypothesis that B has real constant entries is explicit. This theorem does not assert the same basis-cost formula for an arbitrary Hahn-valued constraint matrix, where minors and basis interpolation coefficients can carry additional valuations. For example, take $B = (1, t^\beta)$, $\beta > 0$, and $p = (1, 1)$. Both column bases have entry-cost sum zero, but

$$(H_p)_{11} = \frac{t^{2\beta}}{1 + t^{2\beta}}$$

has valuation 2β , rather than the zero predicted by the formula that ignores the constraint minors.

11 Surreal specialization and preservation of the support group

11.1 No ramification in the relative infinitesimal problem

Corollary 11.1 (Exact local support-group descent). *Let $\Lambda \leq \Gamma$ be the subgroup generated by the supports of all base weights p_e and all perturbations f_e . Then the normalized weights, logarithmic responses, and root-normalized infinitesimal vertex potentials belong to $\mathbb{C}((t^\Lambda))$. No divisible hull, fractional exponent, or larger ordered group is needed for the local normalization.*

Proof. The Hahn field $\mathbb{C}((t^\Lambda))$ contains all the input entries and is closed under the finite field operations defining Π_p . The finite-generator evaluation argument can be carried out inside that same field. It yields h there; restricted exponentiation and multiplication give q there. Solving for the potential on a spanning tree uses only a constant integral inverse matrix. $\square$

This is a local statement. Global scaling over a nondivisible support group can require ramification. For example, over $\mathbb{R}((t^\mathbb{Z}))$ the positive matrix

$$A = \begin{pmatrix} 1 & t \\ 1 & 1 \end{pmatrix} \tag{11.1}$$

cannot be scaled within that field to a bistochastic matrix. Such a matrix would have the form $Q = \begin{pmatrix} x & 1-x \\ 1-x & x \end{pmatrix}$, and invariance of the cross-ratio would give $x^2/(1-x)^2 = t^{-1}$. Thus $2v(x) - 2v(1-x) = -1$, impossible for integer valuations. In the half-integer extension the solution is $x = (1 + t^{1/2})^{-1}$. The contrast with [corollary 11.1](#) is exact: normalization near an already normalized positive-leading base is unramified, even though arbitrary global normalization need not be.

11.2 Transport to surreal and surcomplex numbers

Conway normal form identifies a surreal number with a real series in monomials ω^x with set-sized reverse-well-ordered support; see [\[4, 1\]](#). For a set-sized ordered subgroup $\Lambda \leq (\mathbf{No}, +)$,

the map

$$\sum_{\gamma} a_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} a_{\gamma} \omega^{-\gamma} \quad (11.2)$$

is an ordered field embedding on real coefficients and a field embedding on complex coefficients into $\mathbf{No}[i]$. It preserves normal-form strong sums and the restricted exponential and logarithm. The sign in $\omega^{-\gamma}$ is important: increasing valuation must correspond to decreasing monomial size.

Given finitely many surreal or surcomplex input entries, take the subgroup generated by the negatives of their normal-form exponents. It is a set. Their entire supports lie in that one workspace, and [corollary 11.1](#) keeps the answer there. Therefore all local theorems are genuine theorems about finite surreal and surcomplex matrices, not statements about formal symbols awaiting an interpretation.

For global positive scaling, include a feasible witness positive on every edge of the prescribed support and pass to the divisible hull of the input support group. The resulting real Hahn field is real closed, so [proposition 9.1](#) gives existence there and hence in $\mathbf{No}$. Uniqueness in $\mathbf{No}$ follows by placing any two proposed finite answers in one larger set-sized real-closed workspace. This avoids treating a proper class as an ordinary set-theoretic model.

Order-preserving group embeddings commute with the local construction, because they preserve supports, field operations, and strongly summable evaluation. Complex conjugation also commutes with it:

$$\overline{\mathcal{N}_p(f)} = \mathcal{N}_{\bar{p}}(\bar{f}).$$

In particular, a real base and real perturbation produce a real answer. None of these statements concerns the Berarducci–Mantova scalar derivation; the derivatives implicit in Taylor coefficients are formal derivatives with respect to the edge parameters.

12 Examples, boundaries, and failure modes

12.1 A closed-form square example

Let $b = t^{\gamma}$ with $\gamma > 0$, $a = 1 - b$, and

$$P = \begin{pmatrix} a & b \\ b & a \end{pmatrix}.$$

Multiply only the $(1, 1)$ kernel entry by $1 + z$, where $0 \neq z \in \mathfrak{m}_K$. The infinitesimal square-root branch $r = (1 + z)^{1/2} \in 1 + \mathfrak{m}$ is defined by its binomial series. The normalized answer is

$$Q = \begin{pmatrix} x & 1 - x \\ 1 - x & x \end{pmatrix}, \quad x = \frac{ar}{b + ar}. \quad (12.1)$$

The row and column sums are 1, and $x^2/(1 - x)^2 = (a/b)^2(1 + z)$ is the kernel cross-ratio. All entries are relatively infinitesimally close to their base values, so uniqueness identifies this formula with the constructed branch. The relative errors simplify exactly to

$$\frac{x}{a} - 1 = \frac{b(r - 1)}{b + ar}, \quad \frac{1 - x}{b} - 1 = \frac{a(1 - r)}{b + ar}. \quad (12.2)$$

Thus diagonal entries gain γ in valuation while the two small off-diagonal entries gain zero. The square graph has the same gaps: deleting a diagonal forces a tree of cost 2γ instead of γ , while a minimum tree can avoid either off-diagonal edge. Notice that small normalized entries are not necessarily unstable in *relative* valuation.

12.2 Positivity cannot simply be removed

Example 12.1 (Large formal derivative after phase cancellation). On a square support, take three weights equal to 1 and the fourth $w = -1/3 + t^\beta$, with $\beta > 0$. Then

$$Z = 1 + 3w = 3t^\beta, \quad (H_p)_{44} = \frac{1}{3t^\beta}.$$

All four entry valuations are zero, but a formal derivative coefficient has valuation $-\beta$. The positive-leading assumption fails at w , and cancellation in the tree denominator is exactly the obstruction. This disproves the naive assertion that the same integral-coefficient theorem follows from nonvanishing entries and their valuations alone.

Example 12.2 (Two distinct infinitesimal surcomplex solutions). The failure can be nonlinear, not just a poor derivative bound. Take

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1/3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & -(1/3)\exp s \end{pmatrix}, \quad s = t^{2\delta}, \quad \delta > 0.$$

Every matrix on this support with the same margins as P is of the form

$$Q(z) = \begin{pmatrix} 1+z & 1-z \\ 1-z & -1/3+z \end{pmatrix}.$$

The cross-ratio equation for diagonal equivalence with A is

$$(1 + \frac{1}{3}\exp s)z^2 + \frac{2}{3}(1 - \exp s)z + \frac{1}{3}(\exp s - 1) = 0. \quad (12.3)$$

Its discriminant is exactly $\frac{16}{9}(1 - \exp s)$ and hence has leading term $-(16/9)t^{2\delta}$. It therefore has two square roots in K , obtained from $\pm(4i/3)t^\delta$ times a binomial series in $1 + \mathfrak{m}$. The two roots of (12.3) satisfy

$$z_+ \sim \frac{i}{2}t^\delta, \quad z_- \sim -\frac{i}{2}t^\delta.$$

Both yield matrices relatively infinitesimally close to P , and both are diagonal-equivalent to A . For a full 2×2 support, equality of the nonzero cross-ratios is sufficient: solve three row/column factor equations, and the fourth follows. Here the factors can all be chosen in $1 + \mathfrak{m}$ because all entry ratios are in $1 + \mathfrak{m}$. Thus there are two infinitesimal solutions, and an input change of valuation 2δ produces changes of valuation δ . The uniqueness and no-loss conclusions genuinely fail without an appropriate positivity or noncancellation hypothesis. Positive leading coefficients are a sufficient condition, not claimed to be a necessary one.

12.3 Fixed margins are essential

Consider the tree support of

$$P = \begin{pmatrix} 1 & t^\gamma \\ 0 & 1 \end{pmatrix}, \quad \gamma > 0.$$

Its row margins are $(1 + t^\gamma, 1)$ and its column margins are $(1, 1 + t^\gamma)$. Keep the columns fixed, but change the rows to $(1 + t^\gamma + s, 1 - s)$ with $s = t^\delta$ and $0 < \delta < \gamma$. The unique matrix on this support with the new margins is

$$P' = \begin{pmatrix} 1 & t^\gamma + s \\ 0 & 1 - s \end{pmatrix}.$$

All margin changes are relative infinitesimals, but

$$v(P'_{12}/P_{12} - 1) = \delta - \gamma < 0.$$

Every edge of this graph is a bridge. The exact bridge rigidity of [theorem 6.2](#) holds for *fixed* margins, not for this different problem.

12.4 No topological convergence claim is hidden in the proof

In $\mathbb{Z}_{\text{lex}}^2$, with $\delta = (0, 1)$ and $\beta = (1, 0)$, the strongly defined exponential $\exp(t^\delta) = \sum_{n \geq 0} t^{n\delta}/n!$ is not the limit of its finite Taylor sums in the intrinsic valuation topology. The error after degree D has valuation $(D+1)\delta < \beta$ for every D . The error never enters the neighborhood $\{x : v(x) > \beta\}$. This does not affect existence of the Hahn sum, which is a coefficientwise support statement.

Accordingly, we have proved neither convergence of the alternating Sinkhorn iteration in an arbitrary-rank valuation topology nor convergence of set-indexed sequences in the full surreal fine topology. Our normalization is specified by exact algebraic and strong-series identities. Algorithmic convergence and exact normalization are different assertions.

13 Verification, implementation, and remaining questions

13.1 What the supplied program checks

The package contains `code/verify.py`, using Python and SymPy with exact integers, rational functions, and rational coefficients. It does not use floating-point evidence for a valuation identity. The recorded run used Python 3.13.5, SymPy 1.14.0, and random seed 20260922. It passed the following finite checks:

Check	Recorded scope
Tree interpolation and projections	60 connected bipartite graphs; 348 spanning trees in those graphs; 2,056 projection matrix entries.
Deletion-gap versus minimax path	458 graph/cost instances from 229 connected support graphs; 2,612 edge-gap comparisons, including 1,266 bridges; integer and lexicographic integer-pair costs.
Formal normalization recurrence	Two exact constant-weight examples through degree 7, and two Laurent-weight square examples through degree 6; conservation, cut identities, and coefficient gain bounds.
Chain inverse valuations	Chain sizes $n = 2, 3, 4, 5, 6$, checking 55 inverse entries.
Closed-form square	Three exact rational identities for the cross-ratio and relative errors.

The full machine-readable result is `data/verification.json`; the console record is also included. The graph family consists of all connected spanning subgraphs of $K_{2,2}$, $K_{2,3}$, and $K_{3,3}$, with a fixed-seed sample of 60 used for the more expensive matrix identities. All 229 graphs are used for the two cost-group gap comparisons. Counts include repeated underlying graphs with different cost groups where explicitly stated.

The program writes `data/verification.json` next to its own `code/` directory, so a run replaces the recorded file; it should be run on a copy of this directory. During editorial assembly, such a run under Python 3.14.4 and SymPy 1.14.0 passed in about twenty seconds and reproduced every count above. The file it wrote differed from the recorded one only in the Python version field and in line endings. The main-text proof review reran the unchanged suite on a scratch copy under Python 3.13.14 and SymPy 1.14.0; it passed and matched the historical JSON except for the Python version. The historical files were not overwritten.

These checks are independent corroboration of finite formulas, not a proof of strong Hahn summability, arbitrary-rank existence, or surreal transport. They do not constitute a Lean verification of any new theorem. The general proofs are the arguments in the preceding sections; their imported inputs are identified in the bibliography.

13.2 A reproducible symbolic workflow

Given exact base data p and a perturbation f , the following finite and formal tasks are separated. First form the reduced incidence matrix and the rational projection Π_p . Compute the costs $v(p_e)$ and tree-deletion gains to obtain a precision certificate. Generate the homogeneous terms from (5.5)–(5.6) only to the required total degree. The evaluation lemma supplies the full mathematical denotation; [corollary 6.5](#) supplies the precision of the finite output.

This is an exact-operation specification, not a universal computable representation of all surreal numbers. Arbitrary Hahn coefficients need not come with an effective leading exponent, equality test, or support oracle. The program uses finite graphs and rational-function examples where these questions are decidable by explicit algebra. At higher rank, finite Taylor degree may never reach a requested valuation threshold, as already illustrated. A practical implementation must distinguish a formal exact expression, a support-truncated representation, and a finite Taylor approximation.

A Lean implementation could be modular: prove the incidence-minor and Cauchy–Binet identities over fields; isolate the positive-leading no-cancellation lemma; formalize the triangular coefficient recursion; then connect it to a proved strong-summability evaluation theorem. No assertion is made that these proposed new modules are currently implemented in the repository.

13.3 Questions not settled here

The principal theorems answer the concrete local stability question posed in [section 1](#). They leave other questions open. For a general support graph and two *different* edges, a complete sharp formula for the propagation gain from one prescribed input edge to a prescribed output edge would need to control cancellations between signed tree-path contributions. The chain theorem solves one structured family, not this entire problem.

A second problem is to classify arbitrary surcomplex base weights for which the whole-polydisc branch remains unique and integral. Positive leading coefficients are sufficient, and the examples above show that they cannot simply be omitted. They are not an exact classification of all possible phases or all projection matrices.

Finally, this article does not determine the convergence behavior or iteration complexity of alternating normalization over every higher-rank valued field. Nor does it supply a global surcomplex matrix-scaling theorem for arbitrary complex kernels. These are separate targets; neither is hidden inside the word “analytic” used for our support-controlled local branch.

14 Conclusion

Three results form the core of this article. First, a positive-leading Hahn matrix has a unique surcomplex normalization on the entire relative infinitesimal polydisc, with a finite-generator support proof that works at arbitrary rank. Second, a finite spanning-tree deletion gap is a uniform nonlinear gain in each entry, optimal when $\Gamma \neq \{0\}$, with matching Taylor-coefficient bounds and one-coordinate leading amplitudes. Third, a chain admits explicit nonlinear propagation laws in which successive link valuations add.

The mathematical mechanism is not a general promise that replacing real scalars by surreal scalars preserves analysis. It is a specific combination of finite positive determinant sums, weighted projections, a secant identity, and support-controlled formal algebra. This combination both gives the theorems and states precisely where they can fail. The local answer remains in the exact input support group; the passage to surreal and surcomplex numbers changes the available scales without changing the proof.

A Proof dependency map

Statement	Dependencies
Evaluation lemma	Higman’s word theorem; positive ordered-group sums; finite-generator coefficient algebra.
Tree interpolation	Incidence minors; Cauchy–Binet; Cramer’s rule; positive-leading finite sums.
Whole-polydisc existence	Integral tree projection; triangular formal recursion; evaluation lemma; one-step strict valuation uniqueness.
Sharp nonlinear gain	Existence; exact secant reweighting; rowwise tree formula; no leading cancellation in both tree sums.
Taylor coefficient gain	Formal secant identity; integral inverse of a formal unit; finite tree-weight valuation bounds.
Chain propagation	Existence; cycle basis; exact secant weights; continuant and cofactor identities; strict valuation dominance.
Positive global interpretation	Classical real scaling; real-closed transfer; finite cut bound; local uniqueness.
Surreal specialization	Conway normal form and its Hahn arithmetic; set-sized support-group localization.

Only the global-positive interpretation uses real closedness. Only the last step uses a special property of surreal numbers beyond being realizable as Hahn normal forms. None of the main local proofs uses spherical completeness, Banach contraction, a global surcomplex exponential, or a surreal derivation.

B Build and artifact record

From this directory, build the report with

```
latexmk -pdf -halt-on-error -interaction=nonstopmode article.tex
```

and run the checks on a copy of the directory ([section 13](#)), from the root of that copy, with

```
python -m pip install -r data/requirements.txt
python code/verify.py
```

The bibliography is internal; no external database or figure file is required. Compilation verifies typesetting, not mathematical validity. The delivered PDF was rendered and visually inspected separately from the exact algebra checks.

As delivered, the manuscript was built as `surreal_matrix_scaling.tex`, and its `build.sh`, then at the top of the package, changed to its own directory before running the program and the build. That script is kept unchanged in `code/`, where it changes to `code/` and looks

for `code/verify.py` and `surreal_matrix_scaling.tex` there, neither of which exists. It therefore does not work as placed; use the commands above.

The article, source audit, code, and recorded test outputs belong to this one contribution. The package contained no copies of the repository’s existing reports, and its delivery changed no repository file; [section 1.7](#) records how it was placed.

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